25900293 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECHEA03 - PRODUCTION OPERATIONS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **4** marks

1. Differentiate between thermal cracking and catalytic cracking. [U] / [CO1]
2. Name any four factors that influence hydrocracking efficiency. [K] / [CO2]
3. List any four advantages of catalytic cracking over thermal cracking. [K] / [CO2]
4. Discuss delayed coking and its advantages. [U] / [CO3]
5. What are clathrate hydrates? Mention two problems they cause in petroleum industries. [U] / [CO4]
6. What are acidic and basic oxides? Give one example. [K] / [CO4]
7. Write balanced chemical equations for acrylonitrile production by ammoxidation and identify the major side reaction. [A] / [CO4]

[4x5 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **7** marks

8. Discuss the industrial importance of thermal cracking in the petrochemical industry. Give examples of products obtained and their uses. [U] / [CO1]

9. Discuss the limitations of fixed bed and moving bed processes and how fluidized bed cracking overcomes them. [U] / [CO2]

10. Discuss the desirable properties of commercial hydrocracking catalysts. [U] / [CO2]

11. Explain the manufacture of petroleum coke by thermal cracking. [U] / [CO3]

12. Describe the modified hydroformylation process using rhodium catalyst. How does it improve yield compared to the cobalt catalyst process? [A] / [CO4]

13. Explain the hydrogenation of oxo-aldehydes. Write the reaction and discuss its industrial importance. [A] / [CO4]

14. Explain the role of syngas and aromatic compounds in the petrochemical industry. [U] / [CO1]

[7x5 = 35]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

15. Identify the role of catalytic cracking in modern refinery operations. [K] / [CO2]

16. Explain the principle, types, and chemical reactions involved in reforming. [U] / [CO3]

[15x1 = 15]